

Software Requirements Specification

for

Najah Reminder

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1. Introduction

2. Purpose

The Software Requirements Specification (SRS) provides details regarding the needs of the Najah Reminder system, an integrated notification and reminder system linked to An-Najah National University's Zajel system. The document highlights the system's functionality, emphasizing its ability to provide immediate notifications, reminders, and updates to improve the user experience for students, teachers, and staff members.

3. Document Conventions

This set of rules is gonna be followed during this SRS document

- **Font:** **Bold** for section headings, *italics* for emphasis, and standard text for descriptions.
- **Requirement Identity:** Functional requirements are prefixed with "FREQ" and nonfunctional requirements with "NFREQ".
- **Priority:** Each requirement includes a user priority (1-5, 5 being highest) and technical priority (1-5, 5 being critical for system functionality).

4. Intended Audience and Reading Suggestions

This SRS is intended for:

Developers: To understand system requirements and implementation details (read Sections 3, 4, 5, and 6).

Project Managers: To oversee project scope and priorities (read Sections 1, 2, and 6).

Testers: To verify requirements (read Sections 4, 5, and 6).

University Administration and IT Team: To ensure alignment with university goals (read Sections 1 and 2).

•**Students and Instructors:** To understand system functionality (read Sections 2.2 and 5).

Suggested reading order: Start with Section 1 for an overview, then Section 2 for context, followed by Sections 4 and 5 for detailed requirements and scenarios.

5. Product Scope

Najah Reminder system main goal is to provide real-time notifications, reminders, and updates to Zajel users like (students, instructors, and staff) at An-Najah National University. It provides solutions for the inefficiencies of manual update checks in Zajel, reducing server load and improving user experience. Main objectives include timely notifications, user-friendly interfaces, and integration with Zajel's backend. The system aligns with the university's goal of enhancing academic services. Vision and Scope document is available on Moodle for detailed goals.

6. References

- Vision and Scope Document for Najah Reminder, May 18, 2025.
- Problem Statement and Product Vision, May 18, 2025.
- Najah Reminder Form: [Google Forms Link](#).

7. Overall Description

8. Product Perspective

Najah Reminder is an effective, standalone system integrated with An-Najah National University's Zajel platform. It replaces the manual process of checking updates in Zajel, provided as automated, real-time notifications and reminders. The system interfaces with Zajel's backend to fetch data and supports cross-platform access (web, mobile, SMS). It is not a replacement for Zajel but an enhancement to improve notifications and updates delivery .

9. Product Functions

Real-time Notifications: Deliver updates on exam dates and results, class absence, course registrations, and general announcements.

Reminder System: Send reminders for unchecked notifications, assignment deadlines, exams date, and academic events.

Notification Filtering: Allow users to filter notifications by type and priority (e.g., exams, announcements, dividing sections based on priority).

SMS and Offline Support: Provide notifications via SMS and store critical updates for offline access.

Cross-Platform Access: Support web and mobile platforms.

Feedback and Bug Reporting: Enable users to report issues and provide suggestions.

10. User Classes and Characteristics

Students (Primary, High Importance):

- Frequent users who receive notifications for exam results, course registrations, and deadlines.
- Characteristics: Moderate technical expertise, expect intuitive interfaces, high frequency of use.

Instructors (Secondary, High Importance):

- Receive notifications and may trigger updates (e.g., posting grades).
- Characteristics: Moderate technical expertise, need reliable and timely notifications.

University IT Team (Technical, Moderate Importance):

- Manage system integration and maintenance.
- Characteristics: High technical expertise, focus on security and compatibility.

University Administration (Decision Makers, Low Importance):

- Approve system deployment and ensure alignment with university policies.
- Characteristics: Low technical expertise, focus on user satisfaction and data security.

11. Operating Environment

Hardware: Servers that runs backend processing (for backend), user devices (smartphones, PCs, and Laptops).

Operating Systems: Windows, macOS, iOS, Android.

Software: Compatible with Zajel's backend (assumed REST API), web browsers (Chrome, Firefox, Safari), and mobile apps.

Coexistence: Must integrate seamlessly with Zajel without disrupting its operations.

12. Design and Implementation Constraints

- **Integration:** Must use Zajel's backend and existing API for data access, and improve them to work well with our system.
- **Security:** Must comply with university privacy policies and use encryption for data transmission.
- **Budget:** Limited funding may restrict SMS notifications.
- **Technology:** Use web technologies (HTML, JavaScript, React) for cross-platform support.
- **Scalability:** Must handle high loads during peak periods (e.g., exam seasons, course signing times).

13. User Documentation

- **User Manual:** Online guide for students and instructors using the system.
- **Online Help:** Contextual help within the web and mobile interfaces.
- **Tutorials:** Video tutorials for preparing users, hosted on the university platform.

14. Assumptions and Dependencies

- **Assumptions:**
 - Users prefer push notifications and reminders over manual checks.
 - Zajel's backend provides reliable API access.
- **Dependencies:**
 - Access to Zajel's backend services for data retrieval.
 - University approval for system deployment.
 - Third-party SMS gateway for SMS notifications.

15. External Interfaces and Requirements

16. User Interface

Layout: Clean, intuitive design with a dashboard for notifications and filters.

Standard Elements: Navigation bar, notification list, filter options, feedback Section.

Accessibility: Support for screen readers and keyboard navigation.

Sample Screens:

- **Dashboard:** Displays recent notifications and reminders.
- **Filter Screen:** Allows selection of notification types (e.g., exams, announcements).

- **Standards:** Follow university UI guidelines (provided by the IT team).
- **Error Screen:** Shows a screen with a big red X with an error message, and below describes the error.

17. Software Interfaces

Zajel Backend: REST API (assumed) for fetching notifications and updates, and reminders.

Data: Exam results, course registrations, announcements, absences, deprivations, messages from the counselor.

Format: JSON.

Database: MySQL or cloud Databases for similar for storing user preferences and offline notifications.

Third-Party Libraries: React for web UI, React Native for mobile apps.

SMS Gateway: Twilio or similar for SMS notifications.

18. Communications Interfaces

Protocols: HTTPS for secure API calls, SMS for notifications.

Security: Data encryption (TLS) for all communications.

Standards: REST for API interactions, SMPP for SMS.

19. System Requirements

20. Functional Requirements

21. FREQ-1:

<The User can log in using their Zajel account>

Description: *The User can log in using his Zajel account using registration number and password without the need for sign up.*

User Priority (5/5): *The User can use the Najah Reminder system through Log in and without it the User can't use the system.*

Technical Priority(5/5): This FREQ lets Users use the system, and without it users can't use other system features

22. FREQ-2:

<Real-time Notification Delivery>

Description: The system should fetch and display real-time notifications, like (exam dates, exam results, new Assignments, new course signups, absence and deprivation, and general announcements from the university).

User Priority (5/5): The core feature of the system, and it's the most frequent feature that is gonna be used.

Technical Priority (5/5): This FREQ notifies users when a new message is there, and many features in the system use this FREQ.

23. FREQ-3:

<Notification Filtering>

Description: The system should let the users filter their notifications based on type (exams, grades, assignments, etc), or date (old, new).

User Priority (3/5): Some users like to filter their notifications; they find it easier to filter than searching.

Technical Priority (3/5): Requires Backend logic and Ui.

24. FREQ-4:

<Reminder Notifications>

Description: The system should push reminder notifications to remind the user to check their unchecked notification and incoming deadlines (e.g., Assignments deadlines, exam date, ...).

User Priority (4/5): This FREQ helps users to enhance their academic experience.

Technical Priority (3/5): Requires handling scheduling and backend handling.

25. FREQ-4:

<SMS Notifications>

Description: The system should send notifications via SMS if users link their phone numbers.

User Priority (3/5): (optional feature) Lets the users receive offline notifications via SMS.

Technical Priority (2/5): Depends on budget and third-party services.

26. FREQ-5:

<Offline Notification Storage>

Description: The system shall store critical notifications locally on the user's device for offline access, with a minimum 30-day capacity.

User Priority (3/5): Valuable for connectivity issues.

Technical Priority (3/5): Requires local storage and sync.

27. FREQ-6:

<Feedback and Bug Reporting>

Description: The system should allow users to submit feedback and bug reports via a form.

User Priority (3/5): improves system quality through listening to users.

Technical Priority(3/5): secondary functionality.

28. FREQ-7:

<Display Academic Schedule on Startup>

Description: The system should display a schedule of the student's exams and lecture times on the dashboard immediately after the student logs in to the Najah Reminder application.

User Priority (4/5): Highly valued by students for academic planning, enhances the user experience also.

Technical Priority (4/5): Requires integration with Zajel's API and UI rendering, similar complexity to real-time notifications.

29. FREQ-8:

<Custom Notifications Categories>

Description: The system should allow users to create custom notification categories (e.g., 'Midterm Exams') and assign notifications based on user-defined rules.

User Priority (3/5): Creating custom notification categories enhances user experience by allowing personalized organization of notifications

Technical Priority (3/5): Implementing an intuitive categorization interface requires moderate effort, including UI design and backend rule processing.

30. FREQ-9:

<Notification History Search>

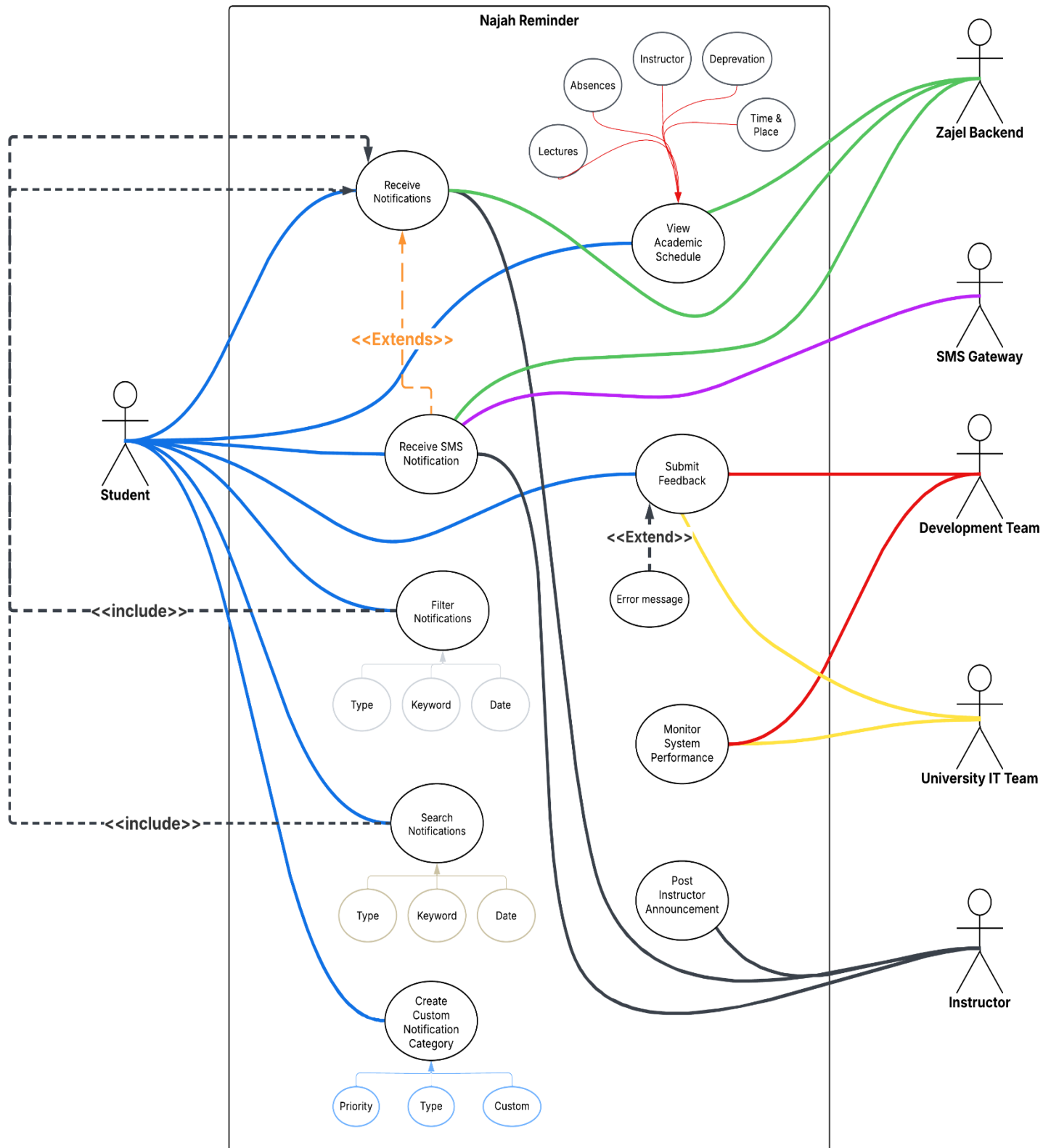
Description: The system should allow users to search the notification history by keywords, date range, or category.

User Priority (3/5): Improves usability for users, and takes less effort to manually search for notifications.

Technical Priority (3/5): Requires backend enhancements techniques to boost the speed of search.

31. System Scenarios

32. Use-case Diagrams



33. Scenarios

34.SCN-1: <Receive and View Exam Result Notification>

FREQ reference	• FREQ-2 (Real-time Notification Delivery) • FREQ-3 (Reminder Notifications) • FREQ-7 (Offline Notification Storage) (if the user has linked his phone number)
NFREQ reference	• NFREQ-1 (Reliability) • NFREQ-3 (Cross-Platform Compatibility) • NFREQ-8 (Accessibility)
Short Description:	A student receives a real-time exam result notification and views its details on his device.
Activation action:	Zajel Backend posts a new exam result, and the system pushes the notification
Precondition:	Student is logged into Najah Reminder with an active Zajel account, and Zajel Backend servers are operating.

Basic flow: <Receive and View Exam Result Notification>		
Step	User Action	System Response
1	None (triggered by Zajel)	Fetches exam results via Zajel API
2	None	Sends push notification to the student's device
3	Student Opens Notifications	Displays exam result details, accessible via screen readers.
4	None	System marks notification as read and stores locally.
Post-condition:	Student is informed of exam result, notification is marked as read, and stored locally for offline access	

35. SCN-2: <User wants to Filter Notifications>

FREQ reference	FREQ-4 (Notification Filtering)
NFREQ reference	- NFREQ-4 (Usability) - NFREQ-8 (Accessibility)
Short Description:	A user filters notifications by type or keywords or specific date.
Activation action:	Zajel Backend sends an API and the system pushes the notification
Precondition:	User is logged into Najah Reminder with an active Zajel account, and Zajel Backend servers are operating.

Basic flow: <User Filters Notifications by Type or date or keyword>

Step	User Action	System Response
1	User selects "Notifications" filter screen	Opens Filter Notifications screen
2	User chooses the type of filter [type, keyword, date]	Queries the database for the required notifications
3	None	System updates screen with the filtered list of notifications
Post-condition:	The wanted notifications are shown on the screen (if found), if not found shows a message with info that the notifications don't exist (Not Found).	

36. SCN-3: <User wants to Filter Notifications>

FREQ reference	FREQ-4 (Notification Filtering)
NFREQ reference	- NFREQ-4 (Usability) - NFREQ-8 (Accessibility)
Short Description:	A user filters notifications by type or keywords or specific date.
Activation action:	Zajel Backend sends an API and the system pushes the notification
Precondition:	User is logged into Najah Reminder with an active Zajel account, and Zajel Backend servers are operating.

Basic flow: <User Filters Notifications by Type or date or keyword>		
Step	User Action	System Response
1	User selects “Notifications” filter screen	Opens Filter Notifications screen
2	User chooses the type of filter [type, keyword, date]	Queries the database for the required notifications
3	None	System updates screen with the filtered list of notifications
Post-condition:	The wanted notifications are shown on the screen (if found), if not found shows a message with info that the notifications don't exist (Not Found).	

37. System Constraints

38. Important Nonfunctional Requirements

39. NFREQ-1:

<Reliability>

Description: The system shall deliver notifications with 99.5% accuracy and within 5 seconds of Zajel updates.

User Priority (5/5): It's critical for users' trust and strong factor to use our system.

Technical Priority (5/5): Requires strong integration with backend, strong bug fixing, and QA.

40. NFREQ-2:

<Security>

Description: The system shall encrypt all data transmissions using strong encryption methods and comply with the university's privacy policies.

User Priority (5/5): It's important to keep users privacy protected, makes the user trust the system.

Technical Priority (4/5): It requires high security protocols and methods to implement the secure system.

41. NFREQ-3:

<Cross-Platform Compatibility>

Description: The system should operate consistently on web browsers (Chrome, Firefox, Safari) and mobile apps (iOS, Android), providing identical functionality.

User Priority (5/5): It lets users access the system via different platforms for accessibility.

Technical Priority (4/5): It requires working on multiple frontends for different devices.

42. NFREQ-4:

<Usability>

Description: The system should be easy to use and interact with for most users.

User Priority (5/5): It lets the user have a great user experience and get the most out of the system, without any kind of suffering.

Technical Priority (4/5): It requires people who master UX/UI, and interact with users to know what most users like and don't like.

43. NFREQ-5:

<Scalability>

Description: The system should handle up to 30,000 users during peak times, with minimal drop-down performance.

User Priority (4/5): It ensures that the system is available for all users during peak times.

Technical Priority (4/5): It requires backend load distribution or balancing, and also optimization.

44. NFREQ-6

<Maintability>

Description: The system should allow updates to core functionalities (e.g., notification logic) with no more than 2 hours of downtime per update.

User Priority (3/5): It affects users when the system is under maintenance, particularly if the period is long, and helps users get bugs fixed quickly and efficiently.

Technical Priority (3/5): It is important for the system to get all bugs fixed with the fastest time available; to let users use all system functionality.

45. NFREQ-7

<Performance>

Description: The system should load the dashboard (including schedule) in under 2 seconds for 99% of user interactions under normal load.

User Priority (4/5): Enhances User-Experience.

Technical Priority (4/5): It helps in testing and developing features, it requires a lot of optimization.

46. NFREQ-8

<Accessibility>

Description: The system shall comply with Modern Accessibility standards, supporting screen readers and keyboard navigation for users with disabilities

User Priority (3/5): It's important for user from all kinds to access and use our system.

Technical Priority (3/5): It requires modern accessibility technologies to include all users.

47. Other Requirements

- **Internationalization:** Support Arabic and English interfaces
- **Database:** Store notification history up to 30 days.
- **Legal:** Comply with university data protection policies.

48. Appendix A: Glossary

- **Zajel:** An-Najah National University's academic management system.
- **Najah Reminder:** Notification and reminder system integrated with Zajel.
- **Notification:** Real-time update (e.g., exam result, announcement).
- **Reminder:** Follow-up alert for unchecked notifications or deadlines.

49. Appendix B: Analysis Models

- Use Case Diagram
- Sequential Diagram

50. Appendix C: To Be Determined List

Exact **Zajel API** specifications (TBD by IT team).

SMS gateway provider and costs (TBD by budget approval).

University policy details for data privacy (TBD by administration).